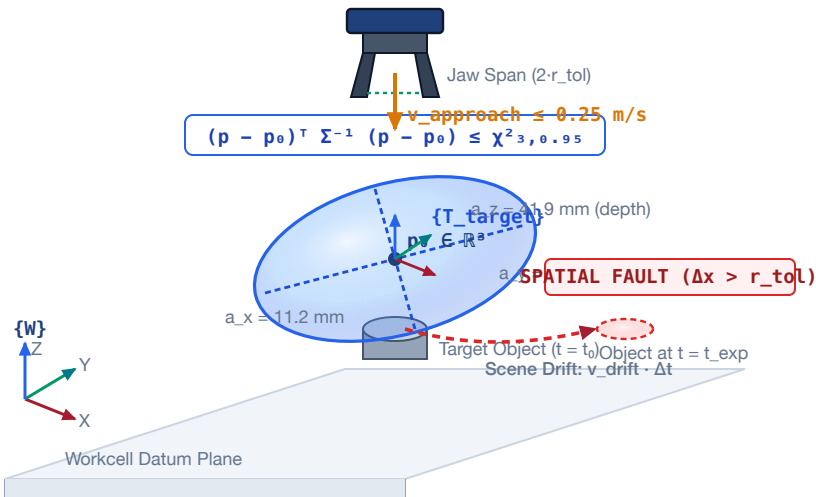


THE INTENT LEASE CONTRACT: SPATIAL TOLERANCE & TEMPORAL EXPIRATION

Bounding Cognitive Intent in Space, Time, and Wrench to Guarantee Safe Termination upon Silence or Failure

SPATIAL CONTRACT · SE(3)

PANEL A: 3D TOLERANCE ENVELOPE



1. Spatial Bound

- **Target Pose:** $T \in SE(3)$
- **Tolerance:** $\Sigma \in \mathbb{R}^{6 \times 6}$
- **Semi-axes:**
 $a = [11.2, 16.8, 41.9]$
- **Volume:** $V = 33.0 \text{ cm}^3$
- **Computable finish test**

2. Temporal Lease

- **Validity:** $\tau \leq 150 \text{ ms}$
- **Expiry:** $t_{exp} = t_{src} + \tau$
- **Refresh:** $T_{period} < \tau$
- **Invariant:**
Silence revokes motion
- **Hardware WWDT Timer**

3. Wrench Bound

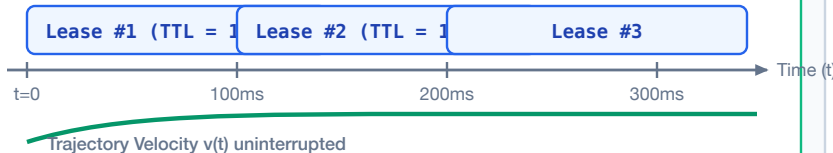
- **Force Cap:** $F_{max} \leq 12 \text{ N}$
- **Torque Cap:** $\tau_{max} \leq 2.5 \text{ Nm}$
- **Power Cap:** $P_{max} \leq 25 \text{ W}$
- **Terminal:**
ACTIVE_HOLD
Prevents runaway torque

TEMPORAL SAFETY · 1000 Hz INVARIANT

PANEL B: MULTI-RATE TIMELINES

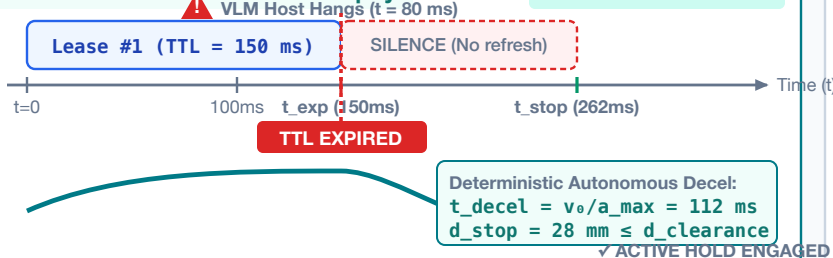
CASE 1: Nominal Lease Renewal Stream

CONTINUOUS C² SPLICE



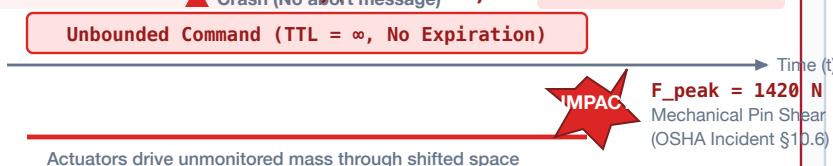
CASE 2: Silent Crash & Local Auto-Expiry

FAIL-SAFE ON SILENCE



CASE 3: Unbounded Goal Runaway (Anti-Pattern)

OPEN-LOOP COLLISION



SYSTEMS INVARIANT: A goal must die on its own.
Silence is safe. Autonomous expiration eliminates reliance on a failed supervisor.